DIPLOMA PROJECT

Development of a Multi-Branch Coffee Shop Management Web Application Using Spring Boot

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Introduction

The hospitality industry is undergoing rapid digital transformation.

Multi-branch coffee shops require centralized, scalable, and secure management systems.

This project proposes a production-ready web solution for digital reservation, menu management, and analytics.

Problem Statement

Many coffee shops still rely on manual processes for reservations and management.

- ! Reservation conflicts and overlapping bookings
- ! Lack of centralized branch management
- ! Limited data analytics
- ! No integrated QR-based ecosystem
- ! Poor customer interaction management

Research and Market Analysis

Existing Systems Analyzed:

- OpenTable
- Toast POS
- Square for Restaurants

Key Observations:

- High subscription costs
- Limited customization
- SaaS dependency
- Restricted data ownership

Identified Research Gap

There is no lightweight, fully customizable, production-ready web application designed specifically for multi-branch coffee shops that:

- Provides centralized control
- Supports branch-level autonomy
- Implements a customizable anti-overlap reservation algorithm
- Integrates QR services into a single ecosystem
- Ensures full data ownership

Project Objective

To design and develop a full-stack web application that:

1. Supports multiple coffee shop branches
2. Provides secure authentication (JWT + Google OAuth)
3. Prevents reservation conflicts
4. Includes analytics and statistics
5. Integrates QR-based functionality
6. Uses modern production-ready architecture

Technologies Used

Backend

Java Spring Boot

Spring Security

(JWT + OAuth2)

Spring Data JPA

Hibernate

Frontend

React

HTML, CSS

JavaScript

Database

PostgreSQL

Other Tools

Swagger (OpenAPI)

Docker

CI/CD Pipeline

System Architecture

Architecture Type: Layered Architecture

Controller Layer — REST API

v

Service Layer — Business Logic

v

Repository Layer — Database Access

Key Features:

- Stateless authentication (JWT)
- Role-based access control
- REST communication
- Dockerized deployment

User Roles

USER

- Register and login
- View menu
- Book tables
- Leave reviews

ADMIN

- Manage tables
- Approve/cancel reservations
- CRUD menu & categories
- Moderate reviews
- View analytics

SUPERADMIN

- Manage branches
- Manage administrators
- System-level configuration

Core Functional Modules

- Authentication Module
- Multi-Branch Management Module
- Reservation Engine (anti-overlap)
- Menu & Category Management
- Review and Rating System
- Analytics Dashboard
- QR Ecosystem Module

Reservation Engine

- Time-slot validation
- Prevention of overlapping reservations
- Reservation editing and cancellation
- Administrator approval workflow
- Capacity-aware booking logic

This ensures operational reliability and fairness.

QR Ecosystem

QR

QR code for each table

Direct menu access

QR

QR code for each branch

Branch information

QR

QR code for campaigns

Promotional content

QR

QR-based review

Customer feedback

Improves customer experience and reduces operational friction.

Database Design (ER Concept)

User

Branch

Table

Reservation

Category

MenuItem

Review

Relationships:

Branch has many Tables | Branch has many MenuItems | User creates Reservations | User writes Reviews | Reservation belongs to Table

Analytics and Reporting

- Daily reservation statistics
- Branch performance comparison
- Reservation trends over time
- Graph-based data visualization

Supports data-driven decision-making.

Minimum Viable Product (MVP)

- v User registration and login
- v Branch-based menu display
- v Online table reservation
- v Reservation approval by admin
- v Basic analytics dashboard
- v QR menu access

Task Plan

Phase 1	Research and System Design
Phase 2	Database and Backend Development
Phase 3	Frontend Development
Phase 4	Integration and Testing
Phase 5	Dockerization and Deployment
Phase 6	Documentation and Final Report

Testing and DevOps

Testing

Unit testing (JUnit)

Integration testing

Documentation

Swagger API documentation

Deployment

Docker containerization

CI/CD pipeline preparation

Value Proposition

For Coffee Shop Owners

- Full control over business data
- Reduced booking conflicts
- Centralized multi-branch management
- Cost-effective alternative to SaaS

For Customers

- Convenient online booking
- Transparent menu and pricing
- Fast QR-based access

Future Work

- Online payment integration
- AI-based demand prediction
- Customer recommendation system
- Mobile application version
- Cloud deployment

References (APA Style)

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